

Noise-Difficulty Indistinguishability: An Information-Theoretic Lower Bound for Data Quality and LLM Hallucination

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1 Theorem Statement

Let $\mathcal{X} \times \mathcal{Y}$ be the input-label space with $|\mathcal{Y}| = K \geq 2$, and $\{f_m\}_{m=1}^M$ be M experts with conditionally independent predictions given x . The ground-truth oracle $f^* : \mathcal{X} \rightarrow \mathcal{Y}$ is unobservable.

Definition 1.1. Two worlds are considered:

- (i) $\mathcal{P}_{\text{noise}}$ (World A): a fraction ρ of inputs suffers *label noise*—the observed label y differs from y^* with probability η_{err} .
- (ii) $\mathcal{P}_{\text{hard}}$ (World B): all $y = y^*$, but the same fraction ρ is *intrinsically difficult*—experts err with effective rate η_{amb} .

Theorem 1.2 (Noise-Difficulty Indistinguishability). *For any $K \geq 2$, $M \geq 1$, there exist $\mathcal{P}_{\text{noise}}$ and $\mathcal{P}_{\text{hard}}$ such that:*

- (1) **Observational equivalence.** $\mathcal{P}_{\text{noise}}(x, y, \{f_m\}) = \mathcal{P}_{\text{hard}}(x, y, \{f_m\})$, iff $\eta_{\text{err}} = \eta_{\text{amb}} = \eta$.
- (2) **Algorithmic lower bound.** For any algorithm $\mathcal{A}$ using only observations,

$$\max(\text{Error}_{\mathcal{P}_{\text{noise}}}(\mathcal{A}), \text{Error}_{\mathcal{P}_{\text{hard}}}(\mathcal{A})) \geq \frac{\eta\rho}{2} > 0. \quad (1)$$

No algorithm can distinguish “label is wrong” from “sample is hard” using observed expert outputs alone—the two worlds are observationally identical, and any discrimination incurs error $\geq \eta\rho/2$.

2 Constructive Proof ($K = 2$)

Fix states s_1, s_2 with $\mathbb{P}(X \in s_1) = \rho$, $\mathbb{P}(X \in s_2) = 1 - \rho$. Choose $\eta \in (0, \frac{1}{2})$, $\varepsilon_1 < \varepsilon_2 \in (0, \frac{1}{2})$.

World A (Noise-driven)

State s_1 : $y^* \equiv 0$. Labels flipped with probability η_{err} : $\mathbb{P}(y \neq y^* \mid s_1) = \eta_{\text{err}}$. Expert accuracy on clean labels: $\mathbb{P}(f_m = y^* \mid s_1, \text{clean}) = 1 - \varepsilon_1$. State s_2 : $y^* \equiv 0$, no noise, accuracy $1 - \varepsilon_2$.

World B (Difficulty-driven)

State s_1 : $y = y^*$ but y^* is random: $\mathbb{P}(y^* = 0 \mid s_1) = 1 - \eta_{\text{amb}}$, $\mathbb{P}(y^* = 1 \mid s_1) = \eta_{\text{amb}}$. Experts are biased toward 0: $\mathbb{P}(f_m = 0 \mid s_1, y^* = 0) = \mathbb{P}(f_m = 0 \mid s_1, y^* = 1) = 1 - \varepsilon_1$. Thus accuracy is $1 - \varepsilon_1$ when $y^* = 0$ but drops to ε_1 when $y^* = 1$. State s_2 : identical to World A.

Equivalence verification

Under $\eta_{\text{err}} = \eta_{\text{amb}} = \eta$, the observable distribution factorises as $\mathcal{P}(x, y, \{f_m\}) = \mathcal{P}(x)\mathcal{P}(y \mid x) \prod_m \mathcal{P}(f_m \mid x)$:

Labels: $\mathcal{P}_{\text{noise}}(y = 0 \mid s_1) = (1 - \eta) \cdot 1 + \eta \cdot 0 = 1 - \eta = \mathcal{P}_{\text{hard}}(y = 0 \mid s_1)$.

Experts: $\mathcal{P}_{\text{noise}}(f_m = 0 \mid s_1) = 1 - \varepsilon_1 = \mathcal{P}_{\text{hard}}(f_m = 0 \mid s_1)$ (since $(1 - \eta)(1 - \varepsilon_1) + \eta(1 - \varepsilon_1) = 1 - \varepsilon_1$).

All marginals match; conditional independence yields identical joint distributions, proving item (??).

Algorithmic impossibility

Let $\mathcal{A}$ flag samples as noisy. Let a be the fraction of $\{x \in s_1, y = 1\}$ flagged. In World A these $\rho\eta$ samples are truly noisy: $\text{Error}_{\text{noise}} = \rho\eta(1 - a)$. In World B they are correct but difficult: $\text{Error}_{\text{hard}} = \rho\eta a$ (false alarms). Hence $\max(\text{Error}_{\text{noise}}, \text{Error}_{\text{hard}}) \geq \frac{\rho\eta(1-a) + \rho\eta a}{2} = \frac{\rho\eta}{2}$, proving (??). Minimax optimum: $a = \frac{1}{2}$ (coin flip). $\square$

Remark 2.1 ($K > 2$). For $K > 2$, construct World B with fully random experts independent of y^* whose marginals match World A’s. This extreme construction—experts as random guessers—suffices as an existence proof. $\square$

3 The Symmetry Condition $\eta_{\text{err}} = \eta_{\text{amb}}$

The condition $\eta_{\text{err}} = \eta_{\text{amb}}$ is *necessary and sufficient* for observational equivalence:

$$\mathcal{P}_{\text{noise}}(y = 1 \mid s_1) = \eta_{\text{err}}, \quad \mathcal{P}_{\text{hard}}(y = 1 \mid s_1) = \eta_{\text{amb}}.$$

When $\eta_{\text{err}} \neq \eta_{\text{amb}}$, the two worlds produce distinct label-class frequencies and become statistically distinguishable—the impossibility collapses. The symmetry thus delineates *precisely when* noise and difficulty are confounded. Interpretation: the probability that flipping noise corrupts a label must equal the probability that ambiguity alone generates an alternative valid label. This is a testable condition: compare label marginals across candidate partitions of $\mathcal{X}$.

4 The Lower Bound $\eta\rho/2$

The bound admits three readings:

- (a) **Information-theoretic.** The two worlds differ on $\rho\eta \cdot n$ samples. Identical observation distributions force coin-flip decisions on these, giving $\rho\eta/2$.
- (b) **Tightness.** Random labelling ($a = \frac{1}{2}$) achieves exactly $\rho\eta/2$, so the bound is minimax-optimal.
- (c) **Positivity.** Strictly positive for any $\eta > 0, \rho > 0$; perfect discrimination requires either zero ambiguity or empty ambiguous subset.

This is a *fundamental* limit—not an artifact of model class or algorithm design.

5 Application: LLM Hallucination

Multi-seed independent decoding

Run an LLM M times on context c with independent random seeds r_m . For fixed model θ , the deterministic forward pass $f_\theta(c)$ is identical; stochastic perturbations $\xi_m = \xi(c; r_m)$ (dropout, sampling, temperature) satisfy $\xi_m \perp \xi_{m'} \mid c$. Error indicators $e_m(c) = \mathbf{1}\{\text{output of seed } m \text{ is wrong}\}$ are conditionally independent: $e_m \perp e_{m'} \mid c$. The M seeds thus constitute M conditionally independent experts, and Theorem ?? applies.

Corollary 5.1 (Hallucination lower bound). *For an LLM with M independent seeds, $\text{Hallucination}_{\text{ambiguous}} \geq \eta\rho/2$, where η is model error probability on ambiguous queries and ρ is their proportion.*

From seed outputs alone, one cannot distinguish model error from essential query ambiguity.

Symmetry in the LLM setting

$\eta_{\text{err}} = \eta_{\text{amb}}$ means: the perturbation ξ_m has equal probability of pushing output away from the correct answer (clear query) as pushing toward a valid alternative (ambiguous query). This depends on logit geometry and is empirically testable.

Breaking the bound

The premise is “only the M seed outputs.” External information breaks it: **RAG** introduces retrieved documents; **RLHF** injects human preference anchors; **tool use** provides execution verification. **Hallucinations cannot be eliminated by scaling alone—only by breaking the information closure of autoregressive generation.**

Acknowledgments

Supported by . Theorem ?? emerged from the SCX axiomatic framework [?] but is fully self-contained: it requires only conditional independence of expert errors.

References

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